

Element C

Presentation and Justification of Solution Design Requirements

Criteria in order of priority:

1. The door can't swing more than 1 degree when locked. $\tan^{-1}(5/30)$. 5 being the door width and 30 inches being around the door's length. (HIGH PRIORITY)
 - a. We found this value to measure the visibility element.
 - b. We weighted this criteria at 40% because it is given in the stimulus that "the child has trouble using the restroom when other people are watching" so this is of very high importance.
2. Must not be opened by <50N of force while locked (HIGH PRIORITY)
Source - Mr. Kiser
 - a. This requirement measures the strength of the lock. Our research shows that the average push on a door is 50N of force, so our lock should be able to withstand this amount.
 - b. We placed this at 25% because the main purpose of the lock is to prevent entry of others.
3. Holds door closed for at least 4 minutes (HIGH PRIORITY)
Source - Mr. Kiser
 - a. This requirement is due to the average time it takes for a child to use the restroom, so this value is sufficient to measure the duration of the lock.
 - b. We weighted this criteria at 15% because the time is less important than the other aspects of success.
4. Fits in the parent's front pocket (approximately, 18 cm in length, 10 cm in width) (HIGH PRIORITY)
Source - Mr. Kiser
 - a. This was given to us by the primary stakeholder out of preference. The compact dimensions make it easily mobile.
 - b. This is weighed at 10% because it is a matter of convenience, however it is a desired factor.
 - c. When reviewing the criteria, the primary stakeholder requested a revision. It was made clear that our dimensions only included two dimensions, rather than three. For this reason, a new conclusion was reached. The pocket was measured to have a depth of 7 centimeters. This parameter is now in effect.

5. Needs to be lighter than 2-3 pounds (HIGH PRIORITY)

Source - Mr. Kiser

- a. This was given to us by the primary stakeholder out of preference. The light weight makes the lock easier to travel with.
- b. This is weighed at 6% because it is a matter of convenience, however it is a desired factor.

6. "Occupied" label that is at least 1-Inch tall (LOW PRIORITY)

- a. This additional requirement was expressed to us by the primary stakeholder. It would give an effective visible warning to the ignorant secondary stakeholders.
- b. This is weighted at 4% due to it having a low effect on the overall success of the lock.

Constraints:

1. Use Materials in class only

Source - Mr. Kiser

2. Complete on January 10th, 2026

Source - Mr. Kiser

3. No sharp metal

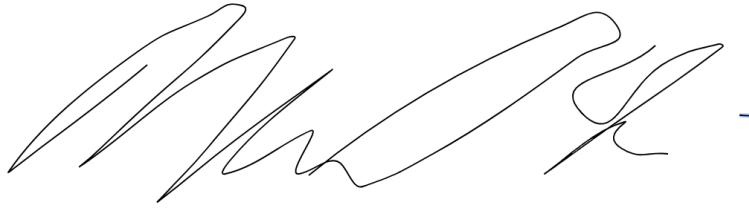
4. No lithium batteries

5. No more than 3oz of liquid

Signature:

Stakeholder Validation of Criteria

Signature:

A handwritten signature in black ink, consisting of several loops and a long horizontal stroke, followed by a small dash.

Citations:

Bathroom Door Thickness

“Toilet Partition Material Comparison Guide: Pros, Cons & Thickness Specs.” *ToiletPartitions.com*, 23 June 2025, <https://accessories.toiletpartitions.com/how-thick-are-bathroom-stall-partitions/>. Accessed 15 Jan. 2026.

Reading Distance

“Understanding Readability Distance for Signs: A Practical Guide.” *Atlas Signs and Plaques*, 26 Oct. 2024, <https://atlassignsandplaques.com/understanding-readability-distance-for-signs-a-practical-guide/>. Accessed 15 Jan. 2026.

TSA Requirements

Transportation Security Administration. *What Can I Bring?* U.S. Department of Homeland Security, <https://www.tsa.gov/travel/security-screening/whatcanibring/all>. Accessed 15 Jan. 2026.